Exact counts and embedding orders for a cubic family

Result. Over every finite field K of cardinality q with q mod3=2, characteristic different from2 and3, and b nonzero, Lean proves the actual elliptic curve $y^2=x^3+b$ is nonsingular and has exactly q+1 points. The point-count trace is0. Every divisor r>2 of its actual cardinality, including composite r, has modular embedding order2. The proof also derives this for every actual point of order>2. [pcc:unverified]

Derived permutation	q mod3=2; e=(2q−1)/3; 3e=2(q−1)+1 Actual cube inverse x^e, including x=0
Actual point count	E_b: $y^2=x^3+b$, char≠2,3 and b≠0 E_b.Point ≃ Option K; #E_b=q+1; trace0
Actual point orders	Every divisor r>2 of #E_b has ord_r(q)=2 Lagrange derives the result for each actual point order

Proof and scope. Derive the inverse cube exponent and its Fermat identities at zero and nonzero elements. For each y there is exactly one x, the cube root of y^2-b. A literal equivalence between actual nonsingular Mathlib points and Option K supplies the cardinality; no count or equivalence is assumed. The discriminant is proved nonzero. Lagrange supplies each actual point-order divisor, and q≡−1 modulo that order gives exact order2 when r>2. For every prime p≥5 with p mod3=2, the specialization derives both characteristic guards; no upper p bound. [pcc:unverified]

Omitted-condition controls. Actual nonsingular-point counts are5 for F5,b=0 (versus predicted6),2 for F2,b=1 (versus3), and12 for F7,b=1 (versus8). Lean directly negates the corresponding coefficient/characteristic/congruence strengthenings. The actual F5,b=1 curve has6points and r=2 divides its count, yet $\mathrm{ord}_2(5)=1$, directly negating exact2. These are omitted-hypothesis controls, not refutations of the qualified family statement. Prior F5 cusp/r2 and the textbook F7 count are credited. [pcc:unverified]

What stays open. This proves the textbook count family and its trace/order consequences. It does not prove the general equivalence between trace, geometric p-torsion, p-rank or endomorphism definitions of supersingularity, general Hasse, canonical full torsion fields or the whole prime-field source theorem. The modular order is distinguished from unproved geometric full-torsion inclusion. The original fourteen-item request remains partial. [pcc:unverified]

Checks and binding. Whole project gate26/0 failures/0 trusted native; only propext, Classical.choice and Quot.sound. Complete explicit1230-character main type hashbb894ca30f297e8548c0a85fdbce66f660ae55191557c09b50f5f846b9c3dcaa; independent native official parser1complete/0prefix/0unresolved. The parser/hash observation is procedural, not a kernel refinement theorem. Six source/config build snapshots, including every failure, are preserved: 69files/32objects/10268-byte lossless XZ archive plus five final gate files. [pcc:unverified]

Sources and reproducibility. Galbraith author chapter9, Exercise9.10.4p184 and Theorem9.11.2/Definition9.11.3/Example9.11.4p189 read; both selected pages visually inspected. PDF SHA f9a3272a…; no fullchapter/bookread or novelty claim. Official Mathlib finite-field Fermat, affine Point equivalences and ZMod order APIs; credited prior CompositeTorsionAudit minus-one argument. KB/Conductor unavailable, primary/prior/official fallback. Lean4.35.0/Mathlib56a00585…, standard defaults; no holes/custom axioms/native proof evaluation/flags/limits/waivers/new crypto experiment/spend/upload. Signed standing remains unverified. Source-linked proof map and pinned sources accompany this report. [pcc:unverified]